

One-Liner

Impact

GDMT

Other
Therapies

Cases

Guideline Directed Medical Therapy for Heart Failure



One-Liner

Impact

GDMT

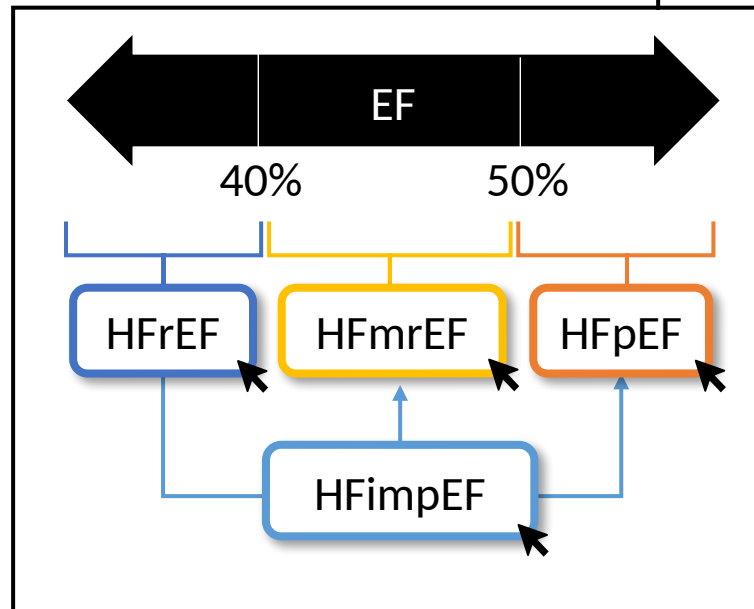
Other
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The Heart Failure One-Liner

Patient with NYHA class ____ heart failure with ____ ejection fraction due to ____ cause?

NYHA	Symptoms
I	None or only with significant exertion
II	Mild: can do below - Climb stairs - Walk 5 mph - Gardening - Heavy housework
III	Mod: can't do above
IV	At rest or with: - Dressing - Bathing - Making the bed



Most common causes



Ischemic

50-60%



Non-ischemic

40-50%

Toxin



Hypertension



Tachycardia-induced



Infectious



Idiopathic/genetic

Others...

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Guideline Directed Medical Therapy for HFrEF

Why We Care: By the Numbers

Number of patients with heart failure in the US?

6 million

1 year mortality if untreated?

25%

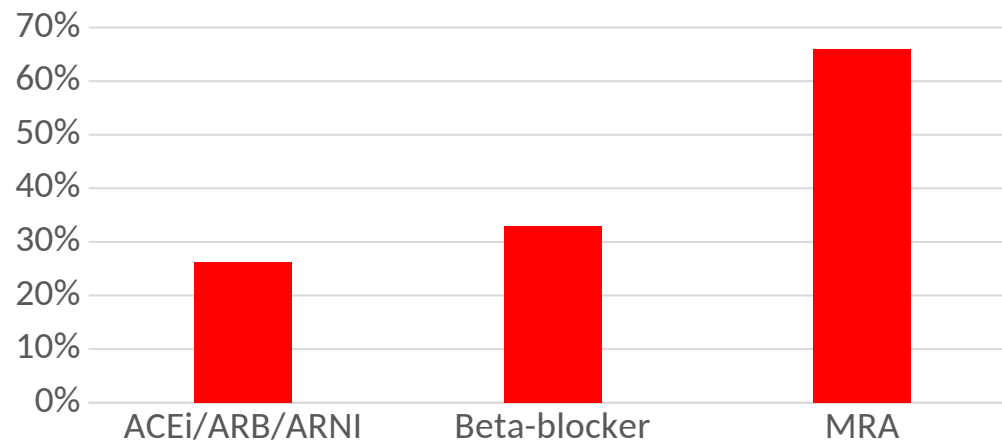
Mortality reduction with GDMT?

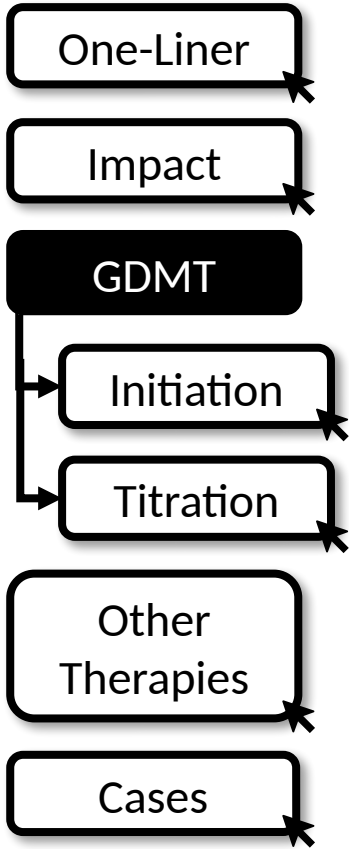
73%

Percent of patients on target doses of GDMT?

1%

% of patients not treated with each class
of medication





GDMT for HFrEF

What medication classes comprise GDMT?

Beta blockers

Metoprolol succinate
Carvedilol
Bisoprolol (less common)

Target doses?
20-50 mg BID

NNT
(3y mortality)

NNT?

MERIT, US CARVEDILOL GROUP, COMET

ARNI

Sacubitril-valsartan

Target doses?

NNT?

PARADIGM-HF (vs. ACE/ARB)

ACEi/ ARB

Lisinopril
Losartan

Target doses?
150 mg daily

NNT?

SOLVD, CONSENSUS

MRA

Spironolactone
Eplerenone (less common)

Target doses?

NNT?

RALES, EMPHASIS-HF

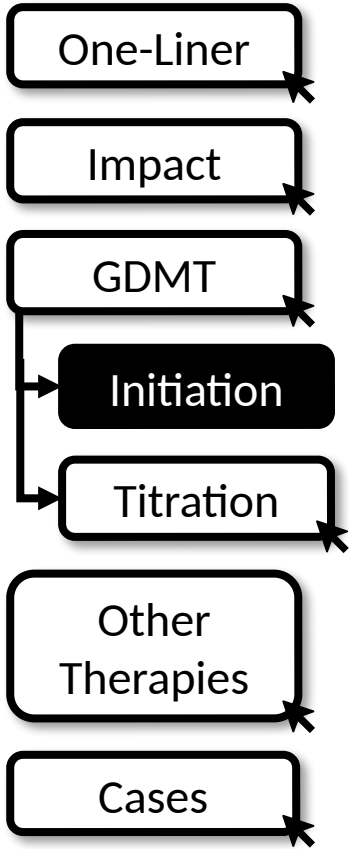
SGLT2i

Dapagliflozin
Empagliflozin

Target doses?
10 mg daily

NNT?

DAPA-HF, EMPEROR-Reduced



GDMT for HFrEF - Initiation

When to initiate?		
Beta blockers	Metoprolol succinate Carvedilol Bisoprolol (less common)	Only when compensated (i.e. euvolemic, no congestion) Negative inotropic, can worsen HF during acute exacerbation
ARNI	Sacubitril-valsartan	Preferred over ACE/ARB When to initiate? No ACEi >36 h
ACEi/ ARB	Lisinopril Losartan	When to initiate? If ARNI is cost prohibitive
MRA	Spironolactone Eplerenone (less common)	Potassium < 5 GFR > 30 When to initiate? (> 2.5 in men) Replacing K supplementation
SGLT2i	Dapagliflozin Empagliflozin	GFR > 20 When to initiate? Caution with frequent UTIs May be cost prohibitive

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Initiation

Titration

Other
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GDMT for HFrEF – Titration & Common Roadblocks

How to adjust GDMT if patient has...

AKI



- Note expected GFR decline with initiation of ACE/ARB/ARNI + SGLT2
- Adjust diuretics
- Reduce ACEi/ARB/ARNI temporarily
- Consider switching ACEi/ARB/ARNI to hydralazine / isosorbide

Hypotension



- Titrate to **symptoms** not only numbers
- Target asymptomatic SBP 90-100
- Avoid non-GDMT medications
- Stagger medication dosing
- Carvedilol to metoprolol succinate

Hyperkalemia

K⁺

- Add SGLT2
- Switch to ARNI from ACE/ARB
- Low K diet
- Consider switching ACEi/ARB/ARNI to hydralazine / isosorbide

Initiation
&
titration

Every 1-2 weeks

Repeat
evaluation
(HR, BP, BMP,
volume status)

Which med to start?

- Goal to start at least low dose of all 4 classes
- Patient dependent, but add/ titrate something!

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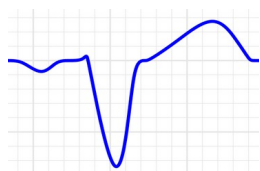
EF Spectrum

Cases

Other Therapies – HFrEF

Persistent EF<35% after at least 3 months of GDMT warrants consideration for additional therapy:

Wide QRS



CRT if QRS >120-150ms, particularly if LBBB

NNT (3y mortality): 8

MIRACLE, COMPANION, MADIT-CRT

Life expectancy
>1 year

Implantable Defibrillator
(SC-ICD or CRT-D)

NNT (3y mortality): 23

MADIT-II, SCD-HeFT

Self-identified
African-
Americans

Hydralazine + isosorbide

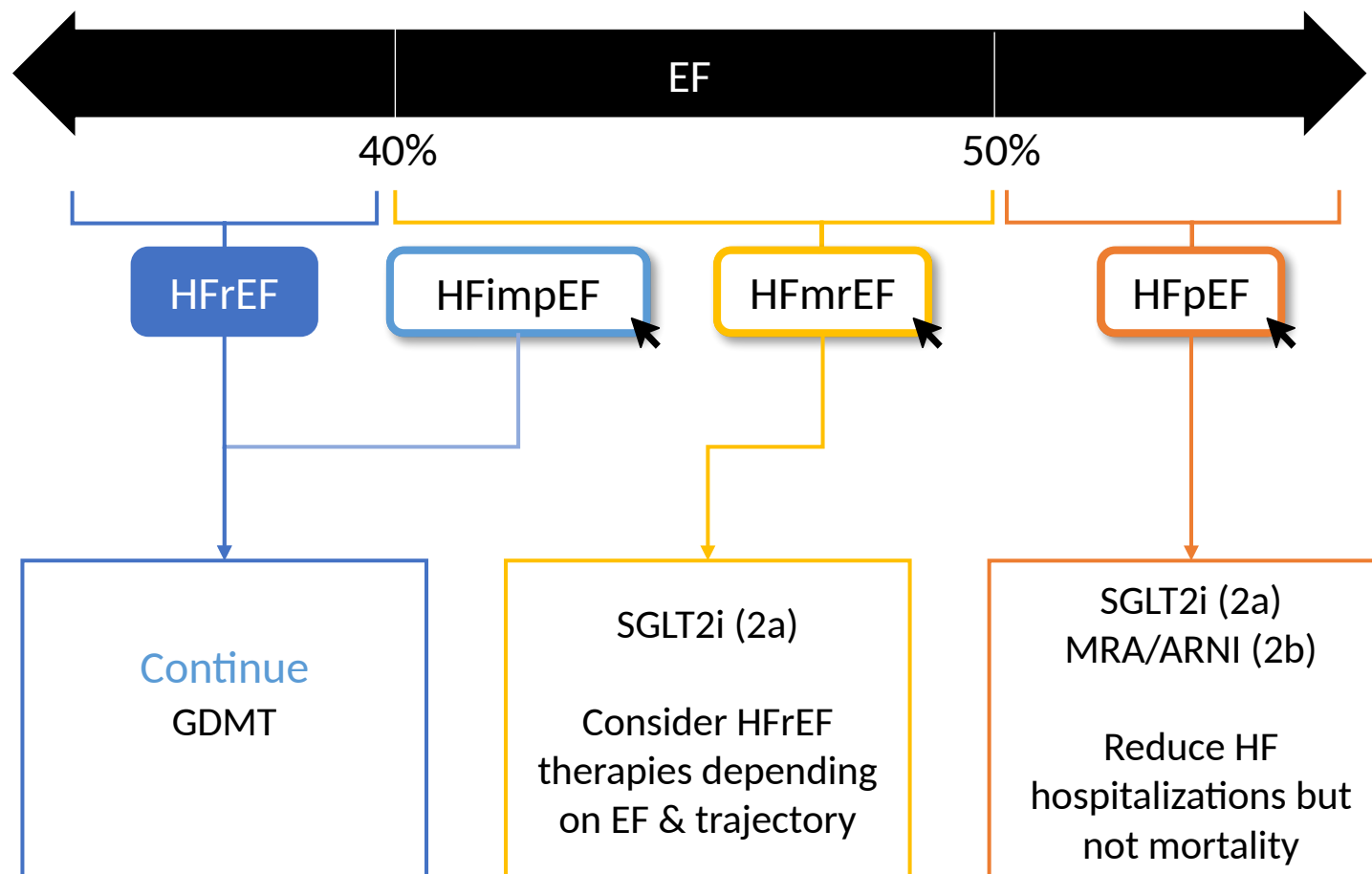
- Class IA rec. for self-identified African-Americans
- Inherent issues with race-based guidelines
- High pill burden



NNT: 8

A-HeFT

Guideline Recommended Therapies – Spectrum of LV Function



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EF Spectrum

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Case 1

63 yo with hypertension, CAD s/p CABG and HF with recurrent admissions for volume overload, here for follow-up.

- Cannot walk to the kitchen or get dressed without shortness of breath
- Meds: aspirin 81 mg, metop tartrate 50 mg BID, lisinopril 40 mg, Lasix 40 mg BID.
- Exam: BP 100/60. Appears euvolemic.
- Labs: Cr 1.3, K 3.2.
- TTE: LVEF 30%, apical/anterior wall motion abnormality.
- ECG: LBBB with QRSd 150 ms.

Heart failure one liner?

NYHA class IV HFrEF (EF <40%), likely **ischemic cardiomyopathy** (focal wall motion abnormalities, CABG)

How can you optimize HF management?

- Transition to ARNI (at least 36h after stopping ACEi)
- Convert to metoprolol succinate 100 mg daily
- Add spironolactone
- Add SGLT2 inhibitor (dapagliflozin, empagliflozin)
- Refer for consideration of CRT-D
- Refer for advanced HF, palliative care

Case 2

67 yo with hypertension, CKD, DM type 2, atrial fibrillation, OSA, and HF.

- Has shortness of breath when walking to the bathroom but not at rest.
- VS: 140/80, HR 90
- Labs: Cr 1.3, K 4.1
- Meds: metop tartrate 50 mg BID, lisinopril 40 mg, metformin 1 g BID, glipizide 5 mg, Lasix 40 daily
- TTE (3 months ago): LVEF of 62%, left ventricular hypertrophy, left atrial enlargement, and grade II diastolic dysfunction.

Heart failure one liner?

NYHA class III HFpEF, likely due to long-standing **hypertension** (LVH)

How can you optimize HF management?

- Start SGLT2 inhibitor, stop glipizide
- Improve BP control, with spironolactone possibly
- Refer for possible rhythm control of AF

Case 3

27 yo with HF and no other past medical history.

- Able to climb several flights of stairs without exercise limitation.
- TTE (3 years ago): LVEF of 25% with global hypokinesis
- TTE (3 months ago): LVEF 50%.
- Labs: Cr 0.8, K 4.2
- VS: BP 110/60, HR 88
- Meds: metoprolol succinate 100 mg, losartan 150 mg

Heart failure one liner?

NYHA class I HFimpEF, likely **non-ischemic cardiomyopathy** (age, global hypokinesis)

How can you optimize HF management?

- Continue current GDMT. No clear benefit to titration for asymptomatic patients with recovered EF.
- Counsel the patient to continue medications indefinitely

Update: HF Guidance Since 2022 (2023–2025)

Original board teaches the 2022 AHA/ACC/HFSA guideline — HFrEF quadruple therapy is unchanged and still correct. New developments since publication:

SGLT2 inhibitors — 2023 ESC focused update made empagliflozin/dapagliflozin Class 1A across HFmrEF and HFpEF (US 2022 guideline lists 2a). SGLT2i now span the entire EF spectrum.

Finerenone (nonsteroidal MRA) — FINEARTS-HF (NEJM 2024) cut CV death + HF events in LVEF $\geq 40\%$. FDA-approved Jul 14 2025 for HF with LVEF $\geq 40\%$; first MRA proven to benefit HFmrEF/HFpEF. Emerging 2nd pillar alongside SGLT2i.

GLP-1 / incretin therapy in HFpEF + obesity — STEP-HFpEF (semaglutide, NEJM 2023) improved symptoms & function; SUMMIT (tirzepatide, NEJM 2025) cut CV death/worsening HF $\sim 38\%$. Treat obesity as a driver of HFpEF.

Teaching point: HFpEF is no longer therapy-orphaned — SGLT2i (1A), finerenone (LVEF $\geq 40\%$), and treating obesity now all have outcome data.

Sources: 2023 ESC Focused Update Eur Heart J 2023;44:3627. Solomon FINEARTS-HF NEJM 2024;391:1475. FDA/Bayer Kerendia approval Jul 2025. Kosiborod STEP-HFpEF NEJM 2023;389:1069. Packer SUMMIT NEJM 2025;392:427.

Take-Home Points

1. Classify every HF patient with a one-liner: NYHA class + ejection fraction (HFrEF ≤ 40 / HFmrEF 41–49 / HFpEF ≥ 50 / HFimpEF) + etiology (ischemic vs non-ischemic).
2. HFrEF gets all four GDMT pillars — beta-blocker, ARNI (or ACEi/ARB), MRA, SGLT2i. Target doses give ~73% mortality reduction, yet most patients are under-dosed.
3. Add or titrate something at every visit. Manage AKI, hypotension, and hyperkalemia rather than stopping GDMT; expect a mild GFR dip when starting ACEi/ARB/ARNI + SGLT2i.
4. Persistent EF $< 35\%$ after ≥ 3 months of GDMT → consider CRT (wide QRS/LBBB) and/or ICD (life expectancy > 1 yr).
5. SGLT2i benefit HFmrEF and HFpEF. Since publication: finerenone is FDA-approved for LVEF $\geq 40\%$ (2025), and treating obesity with GLP-1/tirzepatide improves HFpEF outcomes.

Attribution, License & Optimization Notes

Original lesson: “Chronic Heart Failure Management,” TeachIM.org. Authors: Jonathan Lipsey, MD; Kelsey Flint, MD; section ed. Brandon Fainstad, MD; exec. ed. Yilin Zhang, MD. Published June 2023.

Source URL: teachim.org/teaching_material/chronic-hf/

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Optimization (June 2026): Append-only edit of a copy — original interactive animations, graphics, and slide order preserved. Three slides added: 2023–2025 update, take-home points, this attribution slide.

Currency check: 2022 AHA/ACC/HFSA HFrEF quadruple therapy verified current — no deck errors found. Newer evidence (2023 ESC SGLT2i 1A in HFmrEF/HFpEF; finerenone FINEARTS-HF / FDA 2025; STEP-HFpEF & SUMMIT obesity trials) surfaced on the added update slide and in the HTML package.

Companion file: Chronic_HF_Management_Teaching_Guide.docx (original facilitator teaching script; reviewed, no changes needed).